CSE 4511: Computer Networks

QUIZ 3 **WINTER SEMESTER, 2024-2025**
DURATION: 30 MINUTES **FULL MARKS: 20**

SET: C

STUDENT ID:

---

1. a) Suppose in a custom transport layer protocol named 'AAM', the field to hold the sequence number has 2 bits. What is the minimum and maximum size of the window if it follows the sliding windows technique to send packets? 2 (CO1) (PO1)

Solution:

Minimum - 1, maximum - 3 or $2^2 - 1$

Rubric:

1. For each correct answer, 1 mark

b) Suppose a TCP connection is transferring a file of 5,000 bytes. The first byte is numbered 10,001. What are the sequence numbers for each segment if data are sent in five segments, each carrying 1,000 bytes? 2 (CO1) (PO1)

Solution:

Segment 1: 10,001 Segment 2: 11,001 Segment 3: 12,001 Segment 4: 13,001 Segment 5: 14,001

2. a) Why do we need to prevent the shrinking of the send window in TCP? and how? $2 + 2$ (CO1) (PO2)

Solution:

In Transmission Control Protocol, shrinking of the send window must be prevented because it can create inconsistencies at the sender side, where already transmitted but unacknowledged data may fall outside the newly reduced window. This can lead to confusion, retransmission errors, or violation of reliable delivery guarantees. To avoid this, TCP follows the rule that the receiver should not advertise a smaller window than previously announced. Instead, the window is either kept constant or increased, ensuring stability and correct tracking of in-flight data.

b) How unordered delivery is handled in TCP? 3 (CO1) (PO2)

Solution:

In Transmission Control Protocol, unordered delivery is handled by using sequence numbers and buffering at the receiver. When segments arrive out of order, TCP temporarily stores them in a buffer and waits for the missing segments. Data is only delivered to the application layer once all preceding bytes have been received, thereby ensuring in-order delivery. This mechanism guarantees reliability but may introduce delays if earlier segments are lost.

c) How unordered delivery is handled in SCTP? 3 (CO1) (PO2)

Solution:

In Stream Control Transmission Protocol, unordered data is handled using per-stream sequence numbers (SSN) along with buffering. Within each stream, received chunks are stored and reordered based on SSN, ensuring that data is delivered in sequence for that stream. Unlike TCP, ordering is maintained independently per stream, so missing data in one stream does not block others. When ordered delivery is required, SCTP waits for missing SSNs within that stream before delivering data, thereby reconstructing the correct order efficiently without affecting parallel streams.

3. A UDP segment has the following fields (all values are in decimal): 6 (CO1) (PO2)

- Source IP: 172.16.255.200
- Destination IP: 172.16.255.201
- Protocol: 17
- UDP Length: 15 bytes
- Source Port: 65500
- Destination Port: 65530
- Data bytes: (10, 20, 30, 40, 50, 60, 70)

Calculate the UDP checksum.

Solution:

Given (decimal):

- Source IP: 172.16.255.200
- Destination IP: 172.16.255.201
- Protocol: 17
- UDP Length: 15 bytes
- Source Port: 65500
- Destination Port: 65530
- Data bytes: (10, 20, 30, 40, 50, 60, 70)

Step 1: 16-bit binary words

$$\begin{aligned}
\text{Src IP} &= 1010110000010000,\ 1111111111001000 \\
\text{Dst IP} &= 1010110000010000,\ 1111111111001001 \\
\text{Zero} + \text{Proto} &= 0000000000010001 \\
\text{UDP Length} &= 0000000000001111 \\
\text{Src Port} &= 1111111111011100 \\
\text{Dst Port} &= 1111111111111010 \\
\text{Data} &= 0000101000010100,\ 0001111000101000, \\
&\quad 0011001000111100,\ 0100011000000000
\end{aligned}$$

Step 2: 1's complement addition

$$\text{Final Sum} = 1111100000110011$$

Step 3: Complement 00000111 11001100